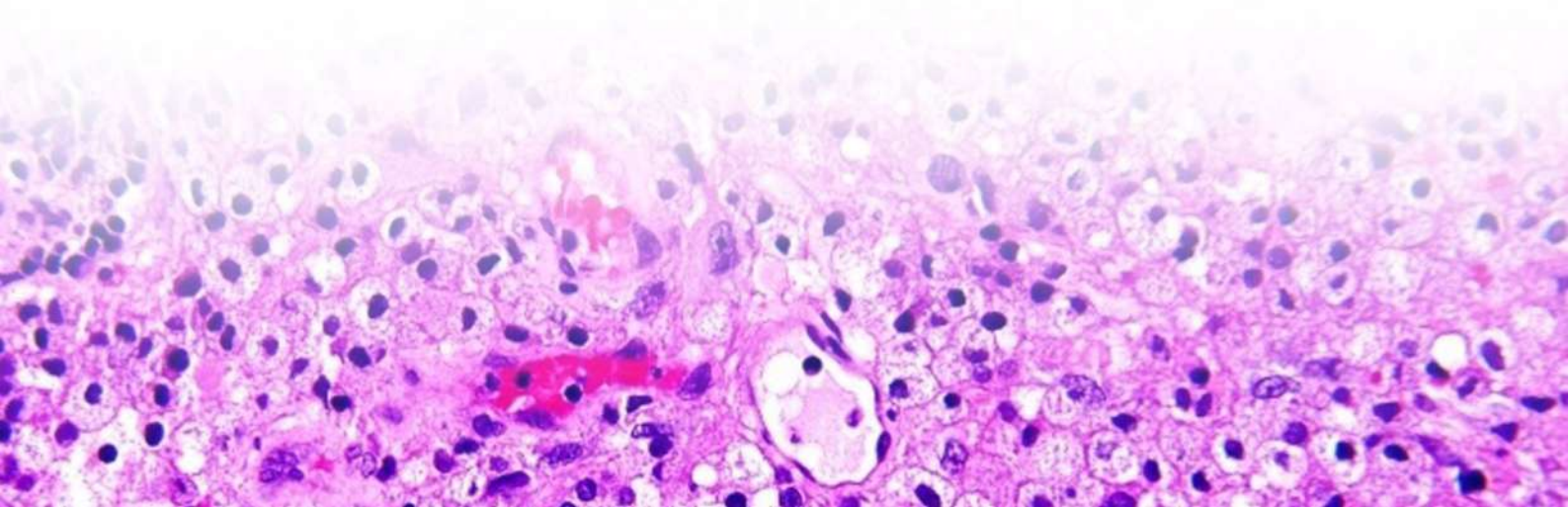
Student Notes

PATHOLOGY

1st Edition

Unit 24

Bone and Skin



Preface

Agam is a group of budding medicos, who are currently doing their under graduation in various Medical Colleges across Tamil Nadu and Pondicherry. The group was initiated on 18th November 2017, in the vision of uniting medicos for various social and professional causes.

We feel delighted to present you Agam Pathology notes prepared by Agam Divide and Rule 2020 Team to guide our fellow medicos to prepare for university examinations.

This is a reference work of 2017 batch medical students from various colleges. The team took effort to refer many books and make them into simple notes. We are not the authors of the following work. The images used in the documents are not copyrighted by us and is obtained from various sources.

Dear readers, we request you to use this material as a reference note, or revision note, or recall notes. Please do not learn the topics for the 1st time from this material, as this contain just the required points, for revision.

Acknowledgement

On behalf of the team, Agam would like to thank all the doctors who taught us Pathology. Agam would like to whole heartedly appreciate and thank everyone who contributed towards the making of this material. A special thanks to Vignesh M, who took the responsibility of leading the team. The following are the name list of the team who worked together, to bring out the material in good form.

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BONES, JOINTS, SOFT TISSUE TUMOURS, PERIPHERAL NERVES AND SKELETAL MUSCLE

ESSAY:

1. Classify tumours of bone. Explain Osteogenic tumours

SHORT NOTES

1. Osteosclerosis
2. Ewing's sarcoma
3. Neurofibroma
4. Osteomyelitis
5. Giant cell tumour
6. Paget's disease of bone
7. Pathogenesis of Osteosarcoma
8. Duchenne Muscular Dystrophy
9. Gout

SHORT ANSWERS:

1. Acoustic neuroma
2. Granular cell tumour
3. Pott's spine
4. Tophi
5. Causes of Osteonecrosis
6. Chondroblastoma
7. Histologic features of giant cell tumour of bone
8. Liposarcoma
9. Radiological appearance of osteosarcoma
10. Pannus
11. Pyogenic Osteomyelitis
12. TB osteomyelitis
13. Aneurysmal Bone cyst
14. Osteochondroma
15. Osteoclastoma
16. Rhabdomyosarcoma

UPDATES

ESSAY:

1. CLASSIFICATION OF BONE TUMOURS

Category and fraction (%)	Behavior	Tumor type	Common locations	Age (yr)	Morphology
Hematopoietic (20)	Malignant	Myeloma Lymphoma	Vertebrae, pelvis	50-60	Malignant plasma cells or lymphocytes replacing marrow space
Cartilage forming (30)	Benign	Osteochondroma Chondroma Chondroblastoma Chondromyxoid fibroma	Metaphysis of long bones Small bones of hands and feet Epiphysis of long bones Tibia, pelvis	10-30 30-50 10-20 20-30	Bony excrescence with cartilage cap Circumscribed hyaline cartilage nodule in medulla Circumscribed, pericellular calcification Collagenous to myxoid matrix, stellate cells
	Malignant	Chondrosarcoma (conventional)	Pelvis, shoulder	40-60	Extends from medulla through cortex into soft tissue, chondrocytes with increased cellularity and atypia
Bone forming (26)	Benign	Osteoid osteoma Osteoblastoma	Metaphysis of long bones Vertebral column	10-20 10-20	Cortical, interlacing microtrabeculae of woven bone Posterior elements of vertebra, histology similar to osteoid osteoma
	Malignant	Osteosarcoma	Metaphysis of distal femur, proximal tibia	10-20	Extends from medulla to lift periosteum, malignant cells producing woven bone
Unknown origin (15)	Benign	Giant cell tumor Aneurysmal bone cyst	Epiphysis of long bones Proximal tibia, distal femur, vertebra	20-40 10-20	Destroys medulla and cortex, sheets of osteoclasts Vertebral body, hemorrhagic spaces separated by cellular, fibrous septae
	Malignant	Ewing sarcoma Adamantinoma	Diaphysis of long bones Tibia	10-20 30-40	Sheets of primitive small round cells Cortical, fibrous , bone matrix with epithelial islands
Notochordal (4)	Malignant	Chordoma	Clivus, sacrum	30-60	Destroys medulla and cortex, foamy cells in myxoid matrix

OSTEOGENIC TUMOURS - Also known as bone forming tumours

OSTEOID OSTEOMA & OSTEOBLASTOMA

Osteoid Osteoma

- <2 cm in diameter.
- Young MEN – 20s.
- Mostly involves femur or tibia, wherein they typically arise in the cortex.
- Severe nocturnal PAIN: PGE2 produced by osteoblasts, Relieved by aspirin (NSAIDs).
- **X-ray findings:** Central Nidus - Reactive cortical bone.
- **Treatment:**
  - NSAIDs
  - Radiofrequency ablation
  - Excision

Osteoblastoma:

- >2 cm in diameter.
- Involves posterior spine (laminae and pedicles) more frequently.
- PAIN – Unresponsive to aspirin.
- No marked bony reaction.
- **Treatment:** Curettage / Excision.

OSTEOID OSTEOMA & OSTEOBLASTOMA

- Benign bone-producing tumours that have identical histologic features but differ in size, sites of origin, and symptoms.
- Round-to-oval masses of hemorrhagic gritty tan tissue.
- Well circumscribed.
- Composed of haphazardly interconnecting trabeculae of woven bone that are rimmed by prominent osteoblasts.
- The intertrabecular spaces are filled by vascularized loose connective tissue.

OSTEOSARCOMA

- Malignant tumour in which the cancerous cells produce osteoid matrix or mineralized bone.
- Most common primary malignant tumour of bone.
- Bulky tumours that are gritty, gray-white, and often contain areas of hemorrhage and cystic degeneration.
- Occurs in all age groups (bimodal age distribution)
  - 75% of people <20 years
  - Older adults, who frequently suffer from conditions known to predispose to osteosarcoma—Paget disease, bone infarcts, and prior radiation.
- Men are more commonly affected than women (1.6 : 1).
- Tumours usually arise in the metaphyseal region of the long bones of the extremities.
- 50% occur near the knee (i.e., distal femur or proximal tibia).
- Present as painful, progressively enlarging masses.
- The neoplastic bone usually has a fine, lace-like architecture but also may be deposited in broad sheets or as primitive trabeculae.
- Tumour cells may produce cartilage or fibrous tissue
- When malignant cartilage is abundant, the tumour is called **chondroblastic osteosarcoma**.
- Radiograph shows a large destructive, mixed lytic and blastic mass with infiltrative margins.

- **Codman triangle:**
  - The tumour frequently breaks through the cortex and lifts the periosteum, resulting in reactive periosteal bone formation.
  - The triangular shadow between the cortex and raised ends of periosteum, known radiographically as Codman triangle.
  - Indicative of an aggressive tumour.
  - The peak incidence of osteosarcoma is during the adolescent growth spurt.
  - Occur most frequently in the growth plate of rapidly growing bones.
  - The formation of bone by the tumour cells is diagnostic.

PATHOGENESIS:

- Germline mutations in **RB** have a 1000-fold increased risk of osteosarcoma and RB mutations are present in up to 70% of sporadic osteosarcomas.
- Patients with **Li-Fraumeni syndrome** (germline TP53 gene mutations) have greatly increased incidence of this tumour.
- **MDM2** and **CDK4** (which inhibit p53 and RB) are overexpressed.
- **CDKN2A/INK4a** (which encodes suppressor genes p16 and p14) is inactivated in many osteosarcomas.

TYPES

Subtypes of osteosarcoma are grouped according to:

- Site of origin (intramedullary, intracortical, or surface).
- Histologic grade (low, high).
- Primary (underlying bone is unremarkable) or secondary to pre-existing disorders (benign tumours, Paget disease, bone infarcts, previous radiation).
- Histologic features (osteoblastic, chondroblastic, fibroblastic, telangiectatic, small cell, and giant cell).

TREATMENT:

- Multimodality approach that includes neoadjuvant chemotherapy: If metastasis is present after Surgery.
- **Prognosis:** 5-year survival rates reaching 60% to 70% in patients without overt metastases at initial diagnosis.
- 10% to 20% of affected individuals have demonstrable pulmonary metastases of those who die of the neoplasm.

SHORT NOTES:

1. OSTEOSCLEROSIS

A disorder that is characterized by abnormal hardening of bone and an elevation in bone density.

TYPES

- **Hereditary osteosclerosis**
  - Malignant infantile osteopetrosis
  - Neuropathic infantile osteopetrosis
  - Infantile osteopetrosis with renal tubular acidosis
  - Infantile osteopetrosis with immunodeficiency
  - IO with leukocyte adhesion deficiency syndrome (LAD-III)
  - Intermediate osteopetrosis
  - Autosomal dominant osteopetrosis **(Albers-Schonberg)**
  - Pyknodysostosis **(osteopetrosis acro-osteolytica)**
  - Osteopoikilosis **(Buschke–Ollendorff syndrome)**
  - Osteopathia striata with cranial sclerosis
  - Mixed sclerosing bone dysplasia
  - Progressive diaphyseal dysplasia **(Camurati–Engelmann disease)**
  - SOST-related sclerosing bone dysplasias
- **Acquired osteosclerosis**
  - Osteogenic bone metastasis caused by carcinoma of prostate and breast
  - Paget's disease of bone
  - Myelofibrosis (primary disorder or secondary to intoxication or malignancy)
  - Osteosclerosing types of chronic osteomyelitis
  - Hypervitaminosis D
  - hypoparathyroidism
  - Schnitzler syndrome
- **Skeletal fluorosis**
  - Monoclonal IgM Kappa cryoglobulinemia
  - Hepatitis C.

2. EWING SARCOMA/PRIMITIVE NEUROECTODERMAL TUMOUR

- Highly aggressive small round cell tumour affecting the diaphysis.
- Uncommon tumours and constitute 6–10% of primary malignant bone tumours.
- **Age:** Mostly seen between 10 and 15 years of age, and about 80% are seen in patients younger than 20 years.
- In children, it is the second most common group of bone sarcomas, first being osteosarcoma.
- **Sex:** More frequent in boys than girls (2:1).

ETIOLOGY AND PATHOGENESIS:

- **Origin:** Multipotent mesenchymal stem cell.
- **Chromosomal abnormalities** – Translocation of EWS gene (present on chromosome 22) with FLI1 gene (present on chromosome 11) creating an **EWSR1/FLI1 fusion gene** (which encodes for EWSR/FLI1fusion protein, that binds to chromatin and dysregulates transcription resulting in uncontrolled proliferation).
- **Site:** Arises in the diaphysis of long tubular bones (Humerus, tibia and femur) and flat bones (Pelvis).

GROSS APPEARANCE:

- Soft, grayish white/tan-white, and frequently show areas of haemorrhage and necrosis.

MICROSCOPY:

- Composed of uniform small round cells with scanty cytoplasm that maybe clear due to glycogen. Absence of septa but presence of scanty stroma
- **Homer Wright Rosettes:** Round cell clusters with central fibrillary core. They indicate neuroectodermal differentiation.

CLINICAL FEATURES:

- Painful enlarged masses, tender, warm and swollen.
- **Systemic features:** It may develop in some and mistaken for infection like osteomyelitis. Fever, weakness, and bone pain seen.
- **Hematological findings:** Anemia, leukocytosis, and raised erythrocyte sedimentation rate.

3. NEUROFIBROMA

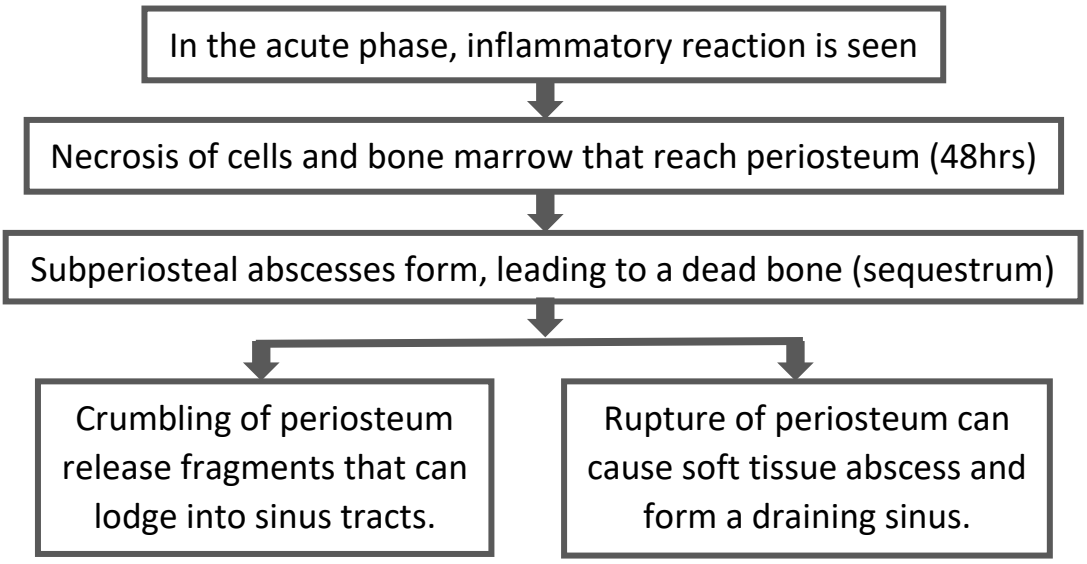
- Benign nerve sheath tumours that are more heterogeneous than schwannomas
- Neoplastic Schwann cells are admixed with perineurial like cells, fibroblasts, mast cells, CD34+ spindle cells.

MORPHOLOGY:

- **Cutaneous**
  - Dermal/ subcutaneous fat, made of spindle cells.
  - Not invasive, but adnexal structures may be enwrapped by lesion.
  - Stroma – collagenized, little myxoid material.
- **Diffuse**
  - Entraps fat & appendages -plaque like appearance.
  - **Pseudo-Meissner corpuscles/tactile like bodies.**
- **Plexiform**
  - Anywhere along a nerve, mostly large nerve trunks involved.
  - Usually multiple, can't separate nerve and lesion.
  - Ropy thickening of multiple nerve fascicles **(bag of worms appearance).**
  - Tumour has ill-defined margins.
  - Loose myxoid material, little cellularity – Schwann cells, inflammatory cells.
  - **Shredded carrot appearance** – collagen bundles.

4. OSTEOMYELITIS

- Osteomyelitis is the inflammation of the bone and marrow that is generally secondary to an underlying infection.
- The changes pertaining to this condition depends on the duration and location of infection.



- In infants, suppurative arthritis is seen which can cause permanent damage.
- Similarly it can also involve vertebrae by causing damage to intervertebral disc.
- Later, chronic inflammatory cells release cytokines that cause reactive bone and fibrous tissue formation → shell of living tissue (involucrum).
    - **Brodie abscess**- introsseus abscess in the cortex.
    - **Sclerosing osteomyelitis of Garre**- involves the jaw.

5. GIANT CELL TUMOURS

- Multinucleated osteoclast-type giant cells randomly and uniformly distributed in a background of mononuclear cells.
- Relatively uncommon, locally aggressive and potentially malignant neoplasm.
- **Age:** Usually seen between 20 and 40 years. In elderly, it may be secondary to irradiation.
- **Cell of origin:** Primitive stromal cells.

PATHOGENESIS:

- The neoplastic cells of giant cell tumour are primitive osteoblast precursors and form only a minority of the tumour cells.
- The major cells of the tumour are non-neoplastic osteoclasts and their precursors. The neoplastic cells express high levels of RANKL.
- They promote the proliferation of osteoclast precursors and their differentiation into mature osteoclasts through RANK expressed by these cells.
- RANK ligand, stimulate the development of surrounding non-neoplastic osteoclast-like cells so osteoclast precursor may also be involved.

MORPHOLOGY:

- Large and red-brown, and often show cystic degeneration.
- Uniform oval mononuclear cells and scattered osteoclast-type giant cells containing 100 or more nuclei
- Mitotic figures are typically frequent.
- Necrosis, haemorrhage, and reactive bone formation also are commonly present.

CLINICAL COURSE:

- GCTs arise in the epiphysis and involve the metaphysis of long bones around the knee (distal femur and proximal tibia), frequently causing pain.

6. PAGET'S DISEASE

- Paget's disease is increased but disordered and structurally unsound bone mass. It presents in 3 stages:
  - **Initial Osteolytic stage**
  - **Mixed osteoclastic-osteoblastic stage** which ends with pre dominance of osteoblastic activity.
  - **Quiescent osteosclerotic stage**.

PATHOGENESIS:

- Mutations resulting in the over activity of NF-KB gene which increases osteoclastic activity**. (SQSTM1 gene)**
- In cases, activating RANK mutation & inactivating OPG mutations.
- Chronic infections of osteoclastic precursors by RNA viruses also contribute to the disease.

MORPHOLOGY:

- **Initial lytic phase**: waves of osteoclastic activity cause resorption pits. Osteoclasts have nuclei more than the normal
- **Mixed phase:** marrow adjacent to the bone formation is replaced with loose connective tissue.
- **Sclerotic phase:** Mosaic pattern of lamellar bone due to unusually prominent cement lines
- All the newly formed bone are remodelled into lamellar bones.
- In the end, the bone has thickened trabeculae & cortices that are soft without the structural stability.

CLINICAL COURSE:

- Symptomatic course and localised pain in the bone is common
- Femur and axial skeleton is commonly affected.
- **Leontisis ossea (lion face):** enlargement of craniofascial skeleton.
- **Platybasia:** invasion of the skull base.
- Severe sec arthritis, chalk stick like fractures, compression fractures of the spine are seen.
- Hypervascularity of the Paget's bone warms up the skin, in polyostotic case they create arteriovenous shunt is formed (cardiac failure).
- Tumours both benign and malignant, mainly sarcoma is seen.

DIAGNOSIS:

- Radiologic findings – Paget bone is a typically enlarged bone with thick coarsened cortices & cancellous bone.
- Increased serum ALP but normal serum calcium & phosphorus.

TREATMENT:

- Mostly with calcitonin & bisphosphonates.

7. OSTEOSARCOMA

- Most common primary malignant tumour that produces osteoid matrix or mineralized bone.
- Bimodal age distribution (more common in Men)
  - Primary – less than 20 years
  - Secondary – Older adults (Paget's disease, bone infarcts and previous radiation – Predisposing factors)
- Most common site involved – Metaphyseal regions of distal Femur and proximal Tibia.

PATHOGENESIS:

- 70% - Complex structural & numerical aberrations and mutations in tumour suppressors and oncogenes.
- Germline mutations in RB gene – Increased risk of Osteosarcoma & Somatic mutation – Sporadic osteosarcoma.
- **Li – Fraumeni syndrome**: germline Tp53 mutation – elevated risk of osteosarcoma.
- **CDKN2A** – inactivated.
- **MDM2 & CDK4** – overexpressed.

MORPHOLOGY:

- **Gross** - Bulky tumour – Gritty, grey white, often contain areas of haemorrhage and cystic degeneration.
- **Diagnostic** – Formation of osteoid matrix or mineralized bone.
- **Microscopy** – Tumour cells are pleomorphic & large hyperchromatic nuclei, bizarre tumour giant cells, vascular invasion and necrosis, tripolar mitosis.

RADIOGRAPHY:

- **Codman triangle** – Triangular shadow between the cortex and raised ends of periosteum – indication of aggressive tumour & not pathognomonic.

TREATMENT:

- Non adjuvant chemotherapy, Surgery, Chemotherapy.

8. DUCHENNE MUSCULAR DYSTROPHY (DMD)

- X linked muscular dystrophy; most common and severe.

PATHOGENESIS

- Xp21 region, encodes for Dystrophin.
- Deletions / Frameshift/ point mutations, sometimes familial.
- Female – carriers, clinically asymptomatic; risk of cardiomyopathy.
- **Dystrophin** - Concentrated over Z bands, link to cytoplasmic actin (extracellular to intracellular transfer of force of contraction).

MORPHOLOGY

- Variation in fibre size.
- Increased internalized nuclei.
- Degeneration, necrosis of muscle fibres and then regeneration.
- Proliferation of endomysial connective tissue.
- Later stages, muscle replaced by fat and connective tissue (Fatty replacement).

CLINICAL COURSE:

- Normal at birth, early motor milestones met on time.
- Walking delayed, 1st sign – clumsiness and inability to keep up with peers.
- Lower leg muscles enlarge with weakness – pseudohypertrophy (bulk converted to fat/ connective tissue).
- May develop arrhythmias/heart failure, cognitive impairment (CNS).
- Death due to respiratory insufficiency, pulmonary infection/cardiac decompensation.

9. GOUT

- Gout is marked by transient attacks of acute arthritis initiated by urate crystals deposited within & around joints.

FEATURES:

- Hyperuricemia (serum uric acid level
- > 6.8mg/dl) due to overproduction or less excretion.
- Recurrent attacks of acute arthritis with monosodium urate crystal in leukocyte of synovial fluid
- Aggregated deposits of monosodium urate crystal in extremities
- Renal disease
- Uric acid nephrolithiasis

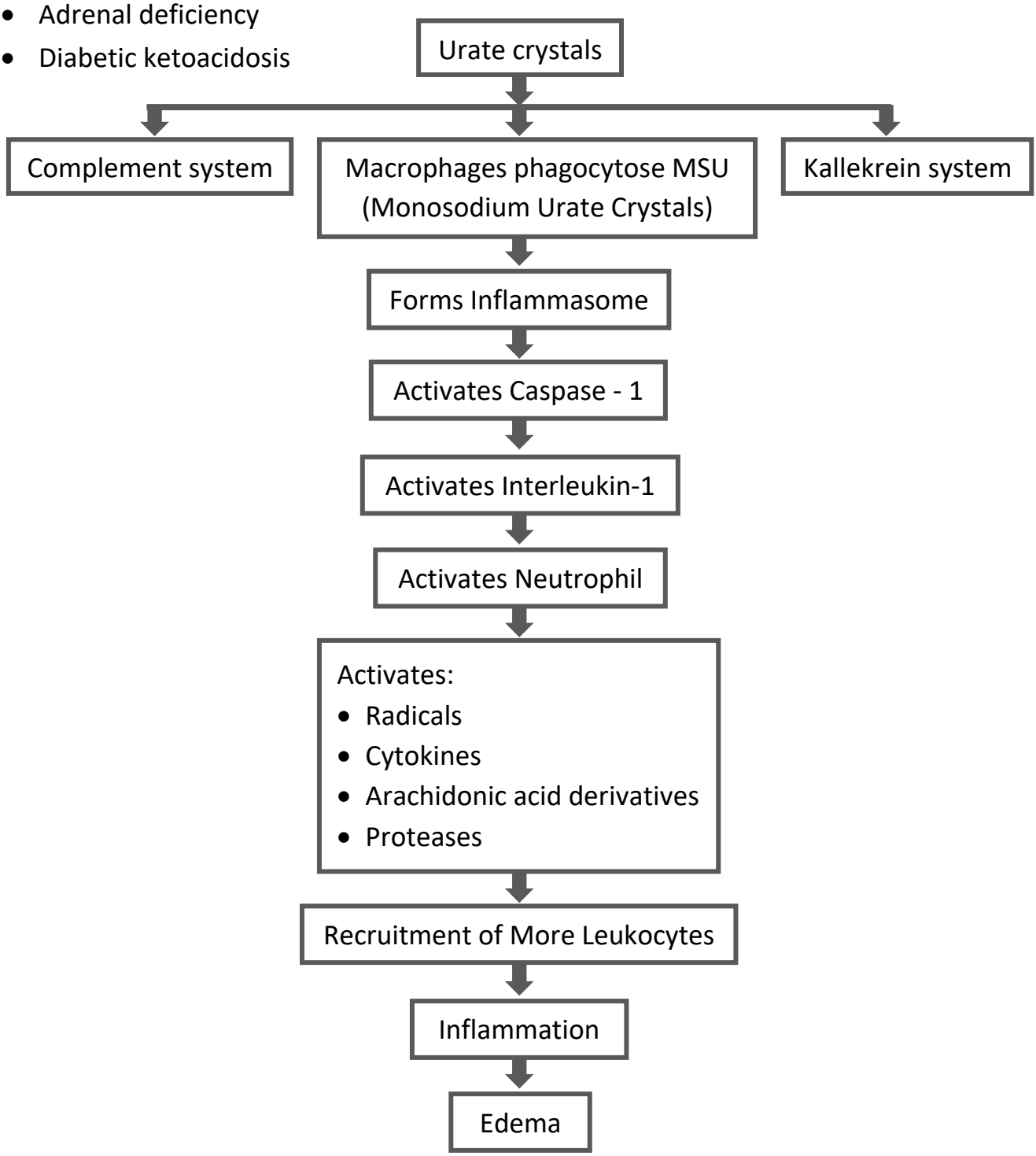
PATHOGENESIS:

Metabolic (90%)

- HGPRTase deficiency (salvage pathway)
- Glucose 6 phosphatase deficiency

Renal (10%)

- Diuretics
- Drugs induced
  - Aspirin
  - Ethanol
  - Nicotinic acid
- Adrenal deficiency
- Diabetic ketoacidosis



FACTORS RESPONSIBLE FOR CONVERSION TO GOUTY ARTHRITIS:

- Age
- Alcohol
- Lead Toxicity
- Genetic Defects
  - HGPRTase
  - URAT-1
  - GLUT-9

CLINICAL HISTORY – FOUR STAGES:

- Asymptomatic hyperuricemia
- Acute gouty arthritis
- Asymptomatic intervals of intercritical periods
- Chronic tophaceous stage

MORPHOLOGY:

- Acute gouty arthritis
- Chronic gouty arthritis – pannus formation, subchondral destruction
- Tophi
  - Located in synovial fluid, periarticular tissue subcutaneously
  - Surrounded ny inflamatory rxn with macrophages, lymphocyte,fibroblasts
- Gouty nephropathy

TREATMENT:

- Lifestyle modification
- Medication to reduce symptoms – NSAIDs
- Lower urate levels – Xanthine Oxidase inhibitor

RENAL MANIFESTATIONS:

- Acute urate nephropathy – by Intra tubular deposits
- Chronic Urate Nephropathy – by Renal Interstitial urate deposits
- Uric acid nephrolithiasis – by Hyperuricosuria caused by Hyperuricemia

PSEUDO GOUT:

- Calcium pyrophosphate crystal deposition disease in more than 50-year-old people.
- Divided into Sporadic(idiopathic), hereditary & secondary types
- Autosomal dominant variant – Germ line mutation in the pyrophosphate transport channel resulting in crystal deposition and arthritis.

SHORT ANSWERS:

1. ACOUSTIC NEUROMA

- Acoustic neuroma is a double neuroma for Vestibular Schwannomas.
- These are benign tumour that exhibit Schwann cell differentiation and often arise directly from peripheral nerves.
- Most Schwannomas cause symptoms by local compression of the involved nerve or adjacent structures.
- They often occur at the CP angle, where they are attached to vestibular branch of eighth nerve and they often present with tinnitus and hearing loss.

2. GRANULAR CELL TUMOR

- Granular cell tumour is a benign tumour of unknown histogenesis.
- Also known as **Abrikossoff's tumour.**
- Tongue is the most common site and next to tongue is subcutaneous tissue in trunk and extremities.
- **Morphology:**
  - **Gross:** Small, firm, grey-white to yellow-tan nodules
  - **Histologically:** It consists of nests or ribbons of large, round or polygonal, uniform cells having finely granular, acidophilic cytoplasm and small dense nuclei.
- The tumours located in the skin are frequently associated with pseudoepitheliomatous hyperplasia of the overlying skin.

3. POTT'S DISEASE

- Pott's disease also known as **Tubercular Spondylitis** is a Mycobacterial osteomyelitis affecting the spine.
- The infection breaks through the intervertebral discs to affect multiple vertebrae and may extend to the soft tissues.
- **Clinical features:**
  - Scoliosis or Kyphosis.
  - Neurologic deficits secondary to spinal cord and nerve compression.

4. TOPHI

- They are the pathognomonic feature of Gout.
- Tophi are formed by aggravates of dissolved Urate crystals surrounded by inflammatory infiltrates, reactive fibroblasts and Giant cells.
- **Sites:** Articular cartilage; ligaments; tendons; bursae, and less frequently soft tissues or kidneys.

5. CAUSES OF OSTEONECROSIS

- **Most common:** Fractures and corticosteroids administration.
- **Other causes:** Alcohol abuse, connective tissue disorders, chronic pancreatitis, infection, trauma, pregnancy, sickle cell crisis and radiation etc.
- All leads to vascular insufficiency through mechanical injury to blood vessels, thromboembolism, external pressure on vessels, or venous occlusion.

6. CHONDROBLASTOMA

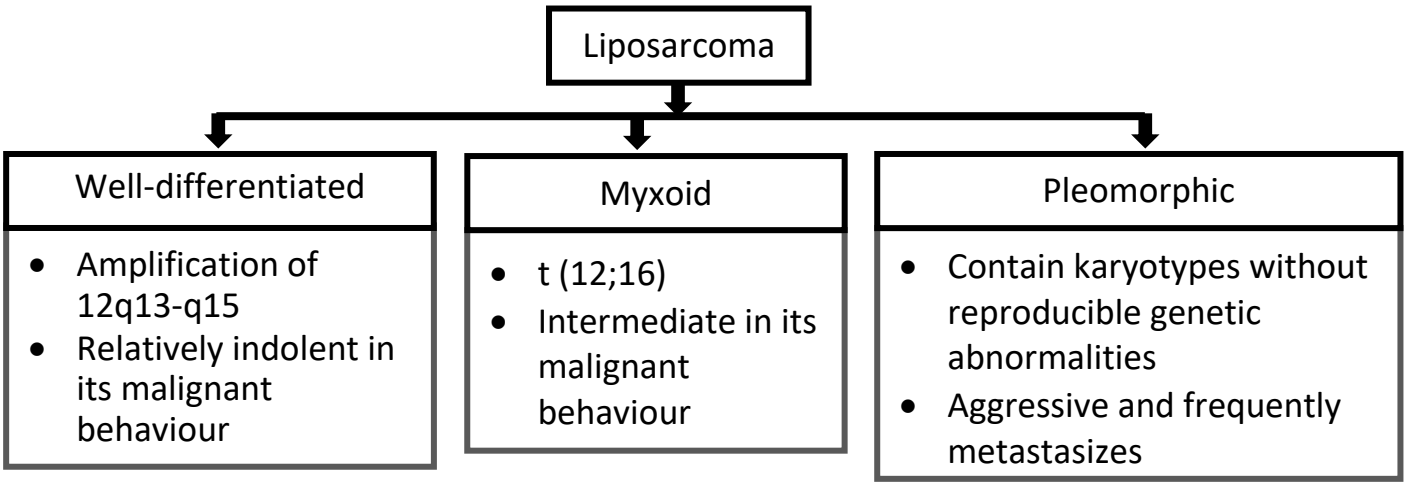
- Benign tumour of bone arising from epiphysis of long bone.
- **Common site:** Upper tibia, lower femur, upper humerus.
- **Common age and sex:** Male at ratio of 2:1 with Female; age:<20 years
- **Gross:**
  - Tumour surrounded by sclerotic bone
  - Cut section: foci of hemorrhage, necrosis, calcification
- **Histology**
  - Chondrocytes-mononuclear, polygonal
  - Osteocyte like giant cells – multi nuclear
  - Cartilaginous intercellular matrix
  - Focal calcification

7. HISTOLOGIC FEATURES OF GIANT CELL TUMOR

- Sheets of uniform oval mononuclear cells and multi-nucleated osteoclast-like Giant cells are present.
- The nuclei of mononuclear cells and osteoclasts are similar.
- Necrosis and mitotic activity may be prominent.

8. LIPOSARCOMA

- Malignant tumour of adipose tissue & most common sarcoma in adults.
- It occurs in the deep soft tissues of the proximal extremities and in the retroperitoneum.

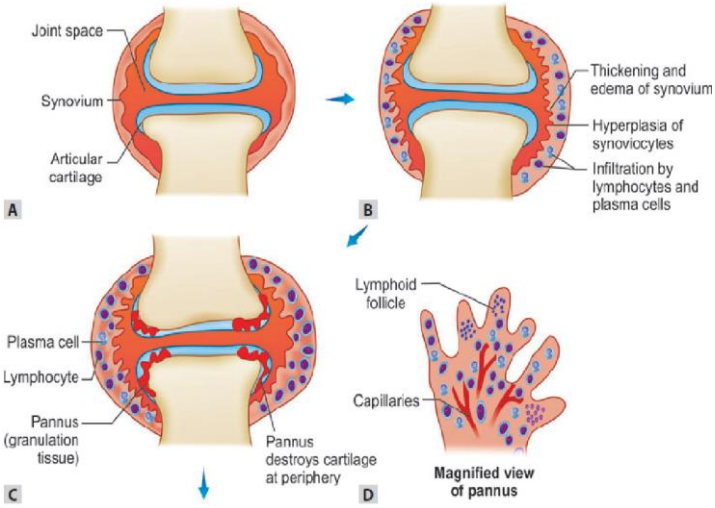


9. RADIOLOGICAL APPEARANCE OF OSTEOSARCOMA

- Characteristic Sunburst appearance due to osteogenesis extending into the soft tissues.
- Presence of Codman triangle – the tumour frequently breaks through cortex and lifts the periosteum. The triangular shadow between the cortex and raised ends of the periosteum.
- The actual lesion is seen as a small round latency that may be centrally mineralized.

10. PANNUS

- A mass of edematous synovium, inflammatory cells, granulation tissue & fibroblasts that grows over the articular cartilage and causes its erosion in rheumatoid arthritis patients.
- The pannus has five features:
  - Hyperplastic synovium
  - Dense inflammatory infiltrates
  - High vascularity
  - Fibro purulent exudate
  - Osteoclastic activity on underlying bone allowing synovium to penetrate joints and to form periarticular lesion and sub chondral cysts



- In advanced untreated cases, the pannus can bridge the bones to form a fibrous ankylosis & later ossify as bony ankylosis.

11. PYOGENIC OSTEOMYELITIS

- Pyogenic osteomyelitis can result from bacteria reaching the bone by:
  - Hematogenous spread (common in children)
  - Extension from a primary site of infection
  - Direct implantation (complication of open fracture)
- Commonly involved bacteria are Staphylococcus aureus, Klebsiella, E.coli.
- They produce cell wall proteins that bind with bone matrix hence facilitating adhesion of bacteria on the bone.
- **Clinical course:**
  - Present as acute systemic illness, fever, chills and throbbing pain & leukocytosis.
  - Radiology presents with lyric focus of bone destruction surrounded by zone of sclerosis.
  - Chronically, complications include pathological fractures, sec. amyloidosis, endocarditis, squamous cell carcinoma.

12. MYCOBACTERIAL OSTEOMYELITIS

- These organisms are usually blood borne from a focus of active visceral disease.
- The bone infection may persist for years and manifests as low grade fever, weight loss, localised pain, fever with chills. They are usually solitary except in immunocompromised patients.
- **Histologic findings:** Caseous necrosis and granulomas that are difficult to control.
- **Tuberculous spondylitis (pott's disease):** Infection involving spine and spread through intervertebral discs to multiple vertebrae and soft tissue.
- **Complication:** Permanent compression fractures, neurological deficits, arthritis, psoas abscess, amyloidosis

13. ANEURYSMAL BONE CYST

- An aneurysmal bone cyst is a benign, blood-filled lesion in the bone that tends to expand or grow.
- True benign bone tumour surrounded by a thin wall of bone.
- Aneurysmal bone cysts (ABCs) can occur in any bone, but are most commonly found around the knee, pelvis or spine.
- ABCs occur slightly more frequently in females than males
- High incidence followed by chondroblastoma and giant cell tumours.
- **Cause:** While the cause of aneurysmal bone cysts is currently unknown, recently ABCs have been linked to a mutation of the ubiquitin specific peptidase 6 (USP6) gene on chromosome 17.
- **Signs and symptoms:** Pain, Swelling, Stiffness, feeling of warmth over the affected area.

14. OSTEOCHONDROMA

- Benign, cartilage capped tumours attached by a bony stalk to the underlying skeleton.
- **Pathogenesis:**
  - Inactivation of EXT1 or EXT2 genes (These tumour suppressor genes encode glycosyltransferase essential for polymerization of heparin Sulfate, an important component of cartilage) through mutation.
  - Develops only in bones of endochondral origin arising at the metaphysis near the growth plate of long tubular bones, especially near the knee; their growth ceases when the normal growth of skeleton is completed.
- **Clinical feature:** Slow-growing mass that can be painful if they impinge on a nerve or if the stalk is fractured.
- **Complications:** Solitary osteochondromas rarely progress to chondrosarcoma or other sarcomas, but malignant transformation occurs more frequently in those with multiple hereditary osteochondromas.

15. OSTEOCLASTOMA/GIANT CELL TUMOUR

- Giant cell tumours are commonly called osteoclastomas due to the mixture of mononuclear cells and multinucleated osteoclast-type giant cells.
- Uncommon, 20-40yr individuals, benign mostly but locally aggressive neoplasm.
- Mononuclear cells in giant cells tumours express RANKL and are believed to be formed via the RANK/RANKL signalling pathway.
- **Morphology:**
  - Large, red brown tumours usually undergo cystic degeneration.
  - Mostly multinuclear cells - proliferate.
  - Scattered osteoclast giant cells.
  - Necrosis, hemosiderin deposition, reactive bone formation.
- **Course:**
  - In adults, it involves both epiphyses and metaphysis; in adolescents, only the metaphysis is involved.
  - Mainly knee, distal femur and proximal tibia – arthritis like symptoms.
  - Exist as solitary tumours.
  - Tumour often erodes into the subchondral bone plate and destroys cortex.

16. RHABDOMYOSARCOMA

- Most common soft tissue sarcoma in childhood & adolescence (<20 years).
- Mostly in head & neck or genitourinary tract or any location where skeletal muscle is present minimally.
- Aggressive neoplasms treated with surgery or chemotherapy with or without radiation.
- Survival depends on location and extent of tumour (**Botryoid subtype of embryonal rhabdomyosarcoma– best prognosis**)

MORPHOLOGY:

- Histologically it is divided into embryonal, alveolar and pleomorphic variants.
- **Rhabdomyoblast** – eccentric eosinophilic granular cytoplasm.

Embryonal	Alveolar	Pleomorphic
● Most common type, soft grey infiltrative mass. ● Sarcoma botryoides, spindle cell and anaplastic variants. ● <10 years, nasal cavity, orbit, middle ear. ● Tumour from mucosa of organ – cambium layer. ● Chromosome 11p 15.5.	● Early-mid adolescence, deep musculature at extremities. ● Network of fibrous septae – look like alveoli. ● Centre cells-discohesive, periphery-adhesive.	● Numerous, large (multinucleated), eosinophilic tumours. ● Deep tissue of adults.

UPDATES

1. OSTEOPETROSIS:

- Albers-Schönberg disease, a mild autosomal dominant form of osteopetrosis, is caused by mutation of CLCN7.
- Most cases of autosomal recessive osteopetrosis are caused by a mutation of TCIRG1, which encodes a subunit of the osteoclast vacuolar H+-ATPase
- Osteopetrosis that is not due to defective acidification may be due to Mutations in IKBKG, which encodes NEMO (X linked & required for osteoclastogenesis and osteoclast survival), the regulatory subunit of the inhibitor of kappaB kinase (IKK) complex that is involved in NF-κB activation.
- IKBKG mutations result in a multisystem disorder called X-linked anhidrotic ecto- dermal dysplasia with immunodeficiency that includes osteopetrosis.

2. ANEURYSMAL BONE CYST

- It associated with USP6-CDH11 fusion gene.
- Radiology: show central lysis and a thin sclerotic “eggshell” of reactive bone at the periphery.
- Supporting trabeculae may create a “soap bubble” appearance on plain films and internal septae with air-fluid levels by computed tomography and magnetic resonance imaging- but all are mostly nonspecific.

3. RHEUMATOID ARTHRITIS:

- HLA-DR4 allele is associated with ACPA-positive RA.
- An epitope on a citrullinated protein, vinculin, mimics an epitope on many microbes and can be presented by the class II HLA-DR4 molecule.
- Long-term treatment with anti-TNF agents predisposes individuals to infection with opportunistic organisms such as M. tuberculosis.

	Osteoarthritis	Rheumatoid Arthritis
Primary pathogenic abnormality	Mechanical injury to articular cartilage	Autoimmunity
Role of inflammation	May be secondary; inflammatory mediators exacerbate cartilage damage	Primary: cartilage destruction is caused by T cells and antibodies reactive with joint antigens
Joints involved	Primarily weight bearing (knees, hips)	Often begins with small joints of fingers; progression leads to multiple joints involved
Pathology	Cartilage degeneration and fragmentation, bone spurs, subchondral cysts; minimal inflammation	Inflammatory pannus invading and destroying cartilage; severe chronic inflammation; joint fusion (ankylosis)
Serum antibodies	None	Various, including ACPA, rheumatoid factor
Involvement of other organs	No	Yes (lungs, heart, other organs)

ACPA, Anti-citrullinated peptide antibody.

4. AXONAL NEUROPATHIES

- Portions of axons that are distal to the point of transection are disconnected from the cell bodies (perikarya) and degenerate
- In cases of toxic and metabolic insults, axons often degenerate in a length-dependent fashion with the longest axons being most susceptible, resulting in a “dyingback” type of pattern of progression.
- The electrophysiologic hallmark of axonal neuropathies is a reduction in signal amplitude owing to the dropout of axons from affected peripheral nerves, with relative preservation of conduction velocity.

5. INFLAMMATORY MYOPATHIES

- Historically, dermatomyositis, polymyositis, and inclusion body myositis have been considered the three main primary inflammatory myopathies:
  - Dermatomyositis as a disease that affects skin as well as skeletal muscles. The histologic hallmark is perifascicular atrophy.
  - Inclusion body myositis, as a slowly progressive disease associated with distinct inclusions termed “rimmed vacuoles” with some similarities to neurodegenerative diseases of the CNS.
  - Polymyositis, to some extent a diagnosis of exclusion, as a T cell–mediated autoimmune disease affecting skeletal muscles that lacks the features of dermatomyositis or inclusion body myositis.
- In addition to the triad mentioned above, other entities have been described and updated classifications have been proposed, but the old historical triad is still used by many physicians.
- In a more granular classification system, many cases traditionally viewed as polymyositis are now regarded as immune-mediated necrotizing myopathy (IMNM), or as a connective tissue disease–associated myositis.
- The systemic connective tissue disorders that can affect skeletal muscle include systemic lupus erythematosus, systemic sclerosis, and sarcoidosis.
- Rarely certain infectious agents can cause inflammation of skeletal muscle.

6. HEREDITARY SENSORY NEUROPATHIES WITH OR WITHOUT AUTONOMIC NEUROPATHY

- This is a diverse group of diseases marked by loss of sensation and variable autonomic disturbances, but with preserved motor strength.
- Loss of pain and temperature sensation is the most common symptom.
- The inability to sense pain leads to traumatic injury of the hands and feet. These are typically axonal neuropathies.

7. NEUROPATHY ASSOCIATED WITH VASCULITIS

- When peripheral neuropathy occurs in systemic vasculitis, it is most common in the MPO-ANCA–associated vasculitides (microscopic polyangiitis and eosinophilic granulomatosis with polyangiitis/Churg-Strauss syndrome) and polyarteritis nodosa and much less common in PR3-ANCA–associated granulomatosis with polyangiitis.
- The most common form of vasculitis associated with peripheral neuropathy is a localized form referred to as nonsystemic vasculitic neuropathy, which is not associated with any anti-neutrophil cytoplasmic antibodies (ANCAs).
- Regardless of the type of vasculitis, peripheral nerves involved by vasculitis typically show patchy axonal degeneration and loss, with some fascicles being more severely affected than others.

8. IMMUNE-MEDIATED NECROTIZING MYOPATHY (IMNM)

- IMNM represents an autoimmune disease that is often associated with distinct autoantibodies (sometimes also referred to as necrotizing autoimmune myopathy).
- Affected patients present with subacute muscle weakness that is typically associated with significantly increased creatine kinase levels.
- Muscle biopsies show fairly prominent myofiber necrosis and regeneration in most cases, while inflammatory cell infiltrates are usually absent or minimal despite the autoimmune nature of the disease.
- In many patients, IMNM is associated with autoantibodies against HMG-CoA reductase.
- Their formation is often attributed to prior statin exposure, but some patients develop these autoantibodies without prior use of these medications. In some patients IMNM has been linked to antibodies to the signal recognition particle

9. INCLUSION BODY MYOSITIS

- Laboratory studies usually show modestly elevated creatine kinase levels
- Most myositis associated autoantibodies are absent
- An antibody to cytosolic 5’-nucleotidase 1A (cN1A) has recently been described.
- It is present in about half of the patients and is a useful marker of the disease.

12. ONE LINERS

- In syphilis, Obliterative endarteritis typically accompanies caseous necrosis as part of gummas.
- Nonhematopoietic Primary Bone Tumors – percentage of cases:
  Cartilage forming (40%)
  Bone forming (32%)
  Unknown origin (19%)
  Notochordal (5%)
- After exclusion of hematopoietic neoplasms, the most common primary bone cancers are osteosarcoma, chondrosarcoma, and Ewing sarcoma.
- Non- specific symptoms such as fatigue and cognitive complaints col- lectively termed posttreatment Lyme disease syndrome.
- KCNQ1 gene polymorphism associated with gout.
- An autosomal dominant variant of pseudo gout (CPPD) is caused by germline gain-of-function mutations in ANKH, an inorganic pyrophosphate transporter, that result in crystal deposition and osteoarthritis at relatively early ages. These mutations are distinct from those associated with craniometaphyseal dysplasia.
- Lipomatosis occurs when multifocal lipomas involve a limb.
- Nodular fascitis is associated with Increased Wnt/β-catenin signaling.
- Spindle cell/sclerosing rhabdomyosarcomas consist of fusiform cells with vesicular chromatin arranged in long fascicles or in a storiform pattern. Rhabdomyoblasts are occasionally present. Dense, collagenous, sclerotic stroma may be more common in adults.
- Caldesmon is a smooth muscle marker.
- The basic morphologic changes that muscular dystrophies elicit in skeletal muscle tissue may differ in severity but do not discriminate between different forms of dystrophy.
- Genes that seemingly stand alone, such as CAPN3 encoding the protease calpain 3 and lamin A/C (which is also mutated in some patients with Emery-Dreifuss muscular dystrophy [EMD];
- The detection of a dystrophin mutation offers definitive diagnosis
- Other patients with anti-CV2 autoantibodies (which recognize CRMP5, an intracellular signaling protein) tend to present with a painful axonal asymmetric sensorimotor neuropathy.
- In some classification schemes myopathies with anti-Jo1 antibodies are described separately as “anti-synthetase syndrome associated myositis.”

- Guillain-Barré syndrome and chronic inflammatory demyelinating polyradiculoneuropathy are the major acute and chronic acquired demyelinating peripheral neuropathies, respectively
- Regardless of cause, patients with disorders that impair the function of neuromuscular junctions complain of painless weakness and fatigue.
- Through interactions affecting the actin cytoskeleton, merlin (product of NF2 gene, mutation causes schwannoma) participates in the regulation of several key signaling pathways that are involved in control of cell shape, cell growth, and the attachment of cells to one another (cell adhesion).
- Patients with peripheral neuropathy often complain of numbness, painful “pins and needles” sensations, and weakness, most often in the distal portions of the extremities

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